

$$W = p\Delta V = 1.0 \times 10^5 (4.0) = 4.0 \times 10^5 \text{ J}$$

- 9 A ball rolls down the circularly curved track shown below.



Which of the following correctly describes its speed, acceleration along the slope and kinetic energy?

	speed	acceleration	kinetic energy
A	increases	increases	increases
B	decreases	decreases	decreases
C	increases	decreases	increases
D	increases	constant	increases

Ans: C

By conservation of energy, as GPE is lost, KE and hence speed is gained by the ball.

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Weight of the ball is constant, but the component of weight along the slope decreases to zero.
Hence acceleration along the slope decreases.

10 A track cycling competition consists of a 4.0 km race and a one hour record race. Cyclists move